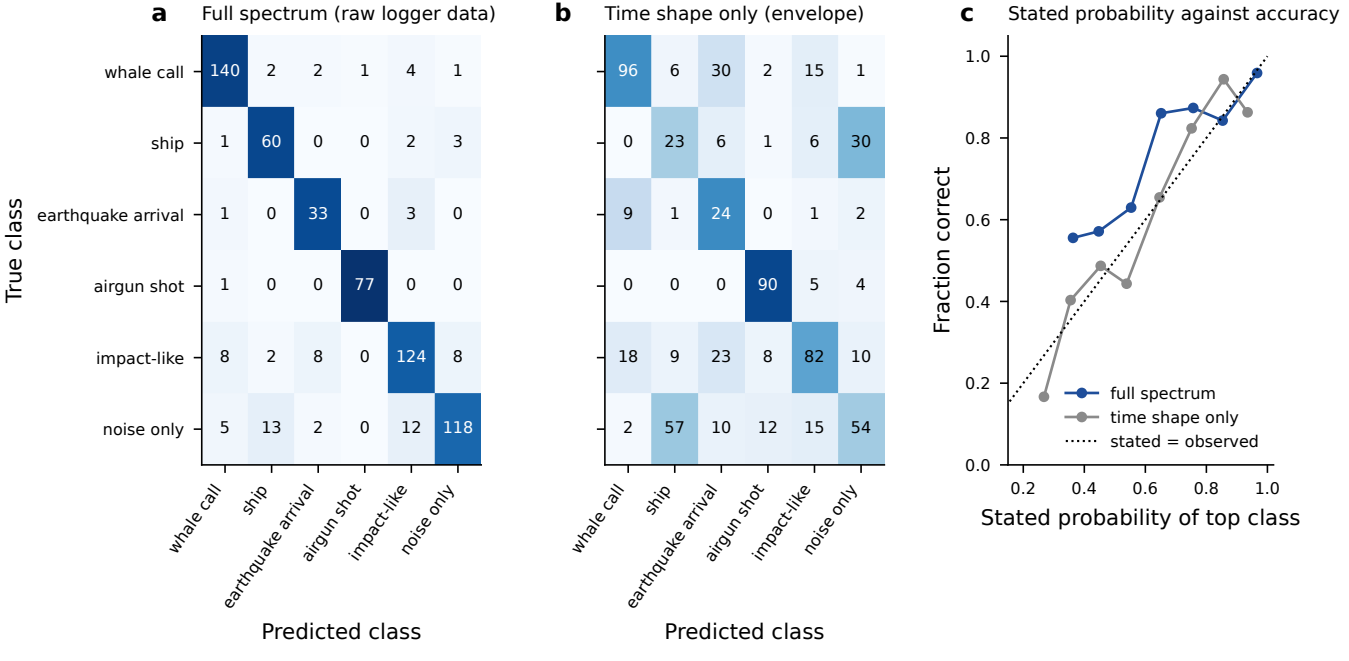
Source classifier: cross-validated confusion and calibration



Counts are leave-one-group-out cross-validated predictions (a group is one catalogued event, or one site and 15-minute slot). Full-spectrum model: 631 examples, 469 groups, Brier score 0.214, calibration error 0.048; it is under-confident below a stated 0.6. Time-shape model: 652 examples, Brier 0.534, calibration error 0.057. No temperature correction was needed (limit 0.10). Training data: IMOS logger records outside every arrival window (same day, and 1-15 March 2014 background days at the same clock hours), labelled by the stated rules of the amendment. Whale, ship and airgun labels come from rules on the same kind of features, so their scores are optimistic. About half of the 37 earthquake labels may be chance timing matches. No ice event passed the labelling rules, so that class is not trained. The impact-like class is synthetic: the loose impact family (slope flat to steep, corner 0.5-20 Hz, fast onset, 0.5-60 s) injected into real noise at 0-12 dB. The time-shape model also holds 21 Diego Garcia South shots from the digitised Kadri 2024 trace. Model: regularised multinomial logistic regression. Class scores never remove a candidate on their own.